INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location	Digital Realty IAD42 (Building R), VA -- station KIAD, America/New_York
Committed pair	OSM 1544360250 -> 1534356804 OSM way 1544360250 / OSM way 1534356804
Facade gap	63.5 m between the two halls
Weather record	43,763 real hourly records from KIAD
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.5180 C at 200 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-09-09 (chatter)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 3 of 24 hours with 1 mode change. That is 2 more than the reactive incumbent, at 0 unsafe hours against its 0.

Agent	3 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	1 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	no	refusal	21.421	21.0	21.110	1.165
01	mechanical	no	dew point	21.114	21.0	20.560	1.075
02	mechanical	no	dew point	21.652	21.0	21.110	1.232
03	mechanical	no	dew point	21.248	21.0	20.560	1.013
04	mechanical	no	dew point	20.541	21.0	19.440	1.056
05	mechanical	no	refusal	20.853	21.0	18.890	1.283
06	mechanical	no	dew point	20.483	21.0	18.890	0.983
07	mechanical	no	dew point	21.533	21.0	19.441	1.587
08	mechanical	no	dew point	23.744	21.0	21.122	2.192
09	mechanical	no	refusal	24.515	21.0	20.560	2.490
10	mechanical	no	dew point	24.645	21.0	20.560	2.267
11	mechanical	no	dry-bulb	25.177	21.0	21.670	1.781
12	mechanical	no	dew point	24.591	21.0	22.780	1.314
13	mechanical	no	dew point	24.435	21.0	23.330	1.258
14	mechanical	no	refusal	25.616	21.0	25.000	1.425

15	mechanical	no	refusal	25.744	21.0	25.000	1.255
16	mechanical	no	refusal	25.811	21.0	26.231	1.190
17	mechanical	no	refusal	26.446	21.0	26.111	1.553
18	mechanical	no	dry-bulb	24.422	21.0	23.332	1.393
19	mechanical	no	dry-bulb	23.535	21.0	21.110	1.425
20	mechanical	no	dry-bulb	22.951	21.0	20.000	1.538
21	FREE	yes	-	20.715	21.0	17.780	1.687
22	FREE	yes	-	20.265	21.0	18.890	1.454
23	FREE	yes	-	18.596	21.0	16.670	1.214

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

01:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 21.114 C would have passed. The outside air is too HUMID: the dew-point bound is 20.28 C against a 15.0 C maximum, failing by 5.28 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

02:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 21.652 C would have passed. The outside air is too HUMID: the dew-point bound is 20.00 C against a 15.0 C maximum, failing by 5.00 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

03:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 21.248 C would have passed. The outside air is too HUMID: the dew-point bound is 19.72 C against a 15.0 C maximum, failing by 4.72 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

04:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 20.541 C would have passed. The outside air is too HUMID: the dew-point bound is 18.61 C against a 15.0 C maximum, failing by 3.61 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

05:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

06:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 20.483 C would have passed. The outside air is too HUMID: the dew-point bound is 18.34 C against a 15.0 C maximum, failing by 3.34 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

07:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 21.533 C would have passed. The outside air is too HUMID: the dew-point bound is 17.77 C against a 15.0 C maximum, failing by 2.77 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

08:00 MECHANICAL [dew point]
Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 23.744 C would have passed. The outside air is too HUMID: the dew-point bound is 18.06 C against a 15.0 C maximum, failing by 3.06 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

09:00 MECHANICAL [refusal]
Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

10:00 MECHANICAL [dew point]
Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 24.645 C would have passed. The outside air is too HUMID: the dew-point bound is 18.89 C against a 15.0 C maximum, failing by 3.89 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

11:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 25.177 C against a 21.0 C limit, so it fails by 4.177 C. It would take a limit 4.177 C higher, or a bound 4.177 C tighter, to change this hour.

12:00 MECHANICAL [dew point]
Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 24.591 C would have passed. The outside air is too HUMID: the dew-point bound is 18.89 C against a 15.0 C maximum, failing by 3.89 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

13:00 MECHANICAL [dew point]
Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 24.435 C would have passed. The outside air is too HUMID: the dew-point bound is 18.89 C against a 15.0 C maximum, failing by 3.89 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

14:00 MECHANICAL [refusal]
Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

15:00 MECHANICAL [refusal]
Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

16:00 MECHANICAL [refusal]
Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

17:00 MECHANICAL [refusal]
Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

18:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 24.422 C against a 21.0 C limit, so it

fails by 3.422 C. It would take a limit 3.422 C higher, or a bound 3.422 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.535 C against a 21.0 C limit, so it fails by 2.535 C. It would take a limit 2.535 C higher, or a bound 2.535 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.951 C against a 21.0 C limit, so it fails by 1.951 C. It would take a limit 1.951 C higher, or a bound 1.951 C tighter, to change this hour.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.715 C, which is 0.285 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.687 C, and the margin is measured, not chosen: 1.527 C of group-conditional forecast error for this hour of day, plus 0.1602 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.265 C, which is 0.735 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.454 C, and the margin is measured, not chosen: 1.189 C of group-conditional forecast error for this hour of day, plus 0.2648 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.596 C, which is 2.404 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.214 C, and the margin is measured, not chosen: 0.953 C of group-conditional forecast error for this hour of day, plus 0.2611 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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